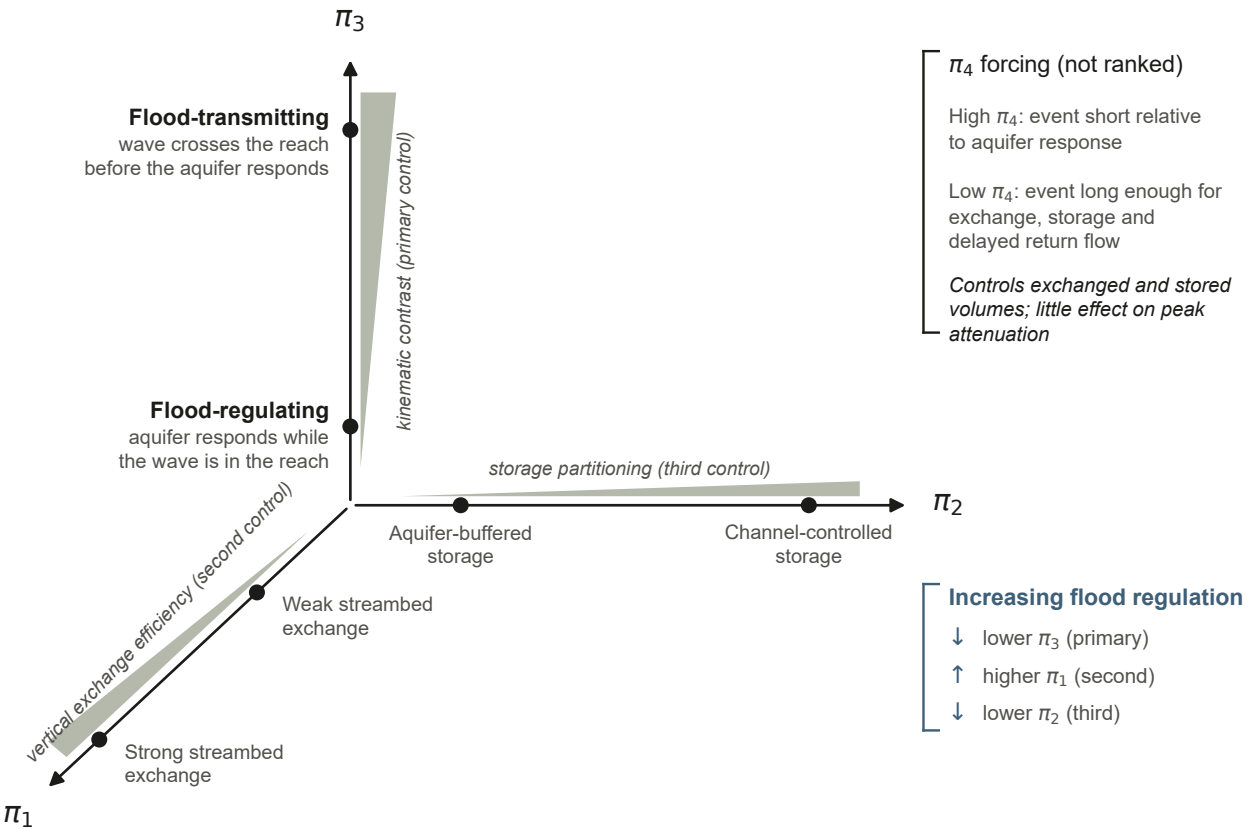

π_3
Flood-transmitting
wave crosses the reach
before the aquifer responds
kinematic contrast (primary control)
Flood-regulating
aquifer responds while
the wave is in the reach
storage partitioning (third control)
π_2
Aquifer-buffered
storage
Channel-controlled
storage
vertical exchange efficiency (second control)
Weak streambed
exchange
Strong streambed
exchange
π_1
π_4 forcing (not ranked)
High π_4: event short relative
to aquifer response
Low π_4: event long enough for
exchange, storage and
delayed return flow
Controls exchanged and stored
volumes; little effect on peak
attenuation
Increasing flood regulation
↓ lower π_3 (primary)
↑ higher π_1 (second)
↓ lower π_2 (third)